Biomedical Engineering

	Normative 4-year Degree
Major	Biomedical Engineering 生物醫學工程
Degree / Award Title	Bachelor of Engineering in Biomedical Engineering 工學士(生物醫學工程)
Offering Academic Unit	Department of Biomedical Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units)

Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12
College-specified Courses	PHY1201 General Physics I	3
	CHEM1200 Discovery in Biology	3

	CHEM1300 Principles of General Chemistry	3

College / School Requirements (0 credit units) — Catalogue Term : Semester A 2025/26

Not required.

Major Requirements (87 credit units) — Catalogue Term : Semester A 2025/26

1. Core Courses (75 credit units)

Course Code	Course Title	Credit Units	Remarks
BME2029	Electrical and Electronic Principles	3	
BME2036	Engineering Computing	3	
BME2066	Professional Engineering Practice	3	
BME2102	Introduction to Biomechanics	3	
BME2103	Medical Biotechnology in Imaging and Measurement	3	
BME2104	Tissue Engineering	3	
BME2105	Introduction to Biomedical Engineering	3	
BME2106	Introduction to Cellular and Biomolecular Engineering	3	
BME2121	Artificial Intelligence in Biomedical Engineering	3	
BME2122	Biological Thermofluids	3	
BME2123	Mathematics for Biomedical Engineering	3	
BME3102	Human Quantitative Physiology	3	
BME3103	Bio-sensors and Bio-devices	3	
BME3104	Robotic Technology in Healthcare	3	
BME3121	Biomedical Signals and Systems	3	
BME3123	Materials for Biomedical Engineering	3	
BME4101	Biomedical Instrumentation	3	
BME4102	Final Year Project	9	A course to fulfil the internship/consultancy project/research project requirement
BME4103	Bio-safety and Risk Assessment	3	
CS1302	Introduction to Computer Programming	3	

CS4465	Computational Biology and Bioinformatics	3	
MA1200	Calculus and Basic Linear Algebra I	3	either MA1200 or MA1300
MA1201	Calculus and Basic Linear Algebra II	3	either MA1201 or MA1301
MA1300	Enhanced Calculus and Linear Algebra I	3	either MA1200 or MA1300
MA1301	Enhanced Calculus and Linear Algebra II	3	either MA1201 or MA1301

2. Electives (12 credit units)

Students are required to earn at least **6 credit units** at B3 and/or B4 levels from the Electives Requirement.

B2 Level Courses

Course Code	Course Title	Credit Units
CHEM2013	Microbiology	3
EE2104	Introduction to Electromagnetics	3

B3 Level Courses

Course Code	Course Title	Credit Units	Remarks
BME3016	Biomedical Engineering CAD	3	
BME3101	Micro and Nanotechnology for Biomedical Engineering	3	
BME3105	Biomedical Systems and Control	3	
BME3122	Fundamental Gene Therapy	3	
BME3200	Internship in Biomedical Engineering	3	A course to fulfil the internship/consultancy project/research project requirement
BMS3101	Cell Transport and Signalling	3	

B4 Level Courses

Course Code	Course Title	Credit Units
BME4032	Robotics and Machine Vision	3
BME4047	Directed Studies	3
BME4104	Technology for Drug Discovery	3
BME4105	Biomedical Devices for Digital Medicine	3
BMS4102	Technology for Regenerative Medicine	3
PHY4232	Radiotherapy Physics	3
PHY4274	Radiation Biophysics	3

3. Optional Courses

Course Code	Course Title	Credit Units
BME3201	Internship in Technology and Engineering	3